That feels convincing: Assessing the quality of pathotic arguments

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Abstract

When an argument involves an appeal to emotion (*pathos*), it is a *pathotic argument*. How can we assess the quality of a pathotic argument? Recent efforts to automatically and computationally identify pathos have produced promising results, however the question of automatically assessing the *quality* of pathotic arguments has been less explored. In this paper, I introduce a theory of pathotic argument quality, and outline how pathotic argument quality could be automatically assessed. I put forward that a high quality pathotic argument should be *effective*, should create *pathotic assent*, and must not be *fallacious*.

Keywords: Argument quality, Computational argumentation, Pathos

1 Introduction

One goal of argumentation theory is to provide methods of assessment that separate good arguments from bad arguments: methods that assess *argument quality*. Broadly, methods for argument quality assessment (AQA) are either concerned with minimal quality (what an argument should minimally achieve or avoid) or maximal quality (what an argument should ideally be) [30]. Minimal quality assessment can involve the use of critical questions [34] or other methods for identifying fallacies. Maximal quality might focus more on how the persuasiveness and convincingness of an argument can be improved.

AQA can now be performed automatically, and automatic AQA has numerous potential applications. Automated essay scoring, for example, checks whether arguments in the essays are sufficiently supported [28]. Search and retrieval systems can

I would like to thank Jodi Schneider, Christoph Stade, and Athena Hatt for invaluable feedback and support. Additionally, thank you to the anonymous reviewer and everyone at the ECA 2025 summer school for their feedback and encouragement. Finally, thank you to the University of Wisconsin-Madison Information School for funding my travel to ECA 2025.

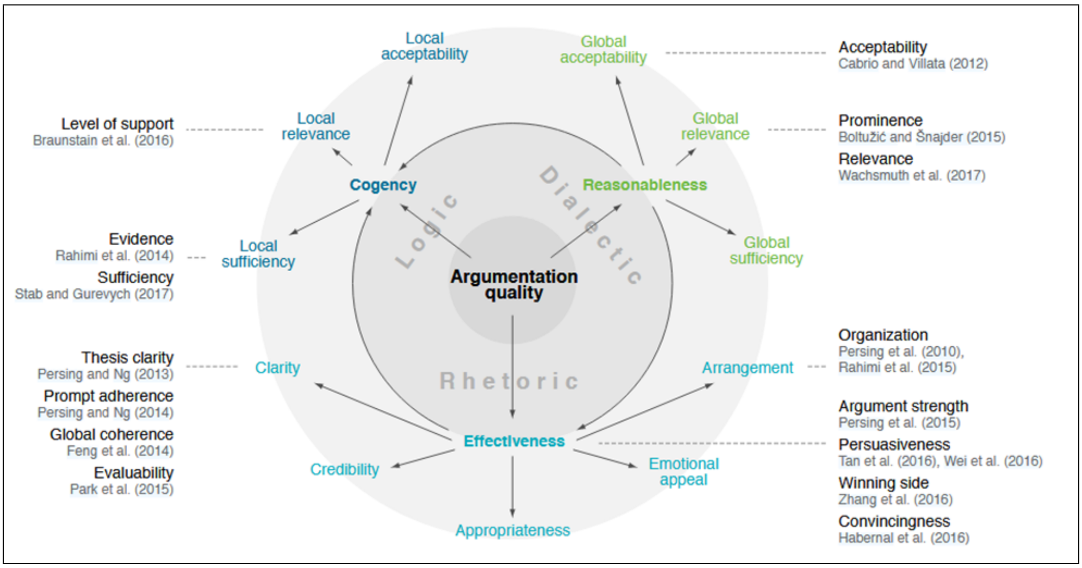

Figure 1: AQ taxonomy from [31]. Pathos-based dimensions are emotional appeal and appropriateness. Licensed under the Creative Commons Attribution 4.0 International License (CC BY 4.0), `https://creativecommons.org/licenses/by/4.0/`

rank results by argument relevance [32]. In online debate, AQA could be used to help participants write more persuasive arguments and find flaws in their opponents' arguments [27].

It is valuable to draw on theoretical concepts of argument quality when designing methods for automatic AQA. A challenge for a theoretically-grounded approach to automatic AQA is that theoretical concepts of argument quality are disparate. For example, the assessment of an argument's clarity is distinct from the assessment of its relevance, but both contribute to or have an effect on an argument's quality: they are examples of *dimensions* of argument quality. One option is to focus on just one dimension of argument quality, for example developing automatic methods for the assessment of sufficiency [28], relevance [32], or convincingness [14, 15]. Alternatively, the interdependencies of a broader set of argument quality dimensions can be analyzed. The argument quality taxonomy (AQ taxonomy) from Wachsmuth et al. (2017b)[31] is a foundational example of a multidimensional approach to automatic AQA. The major contribution of the AQ taxonomy is that it unifies previously disparate theoretical views by establishing a common ground for work on AQA. In the AQ taxonomy, argument quality can be analyzed as a loose and overlapping taxonomy that has rhetorical, logical, and dialectical dimensions (see Figure 1). Ar-

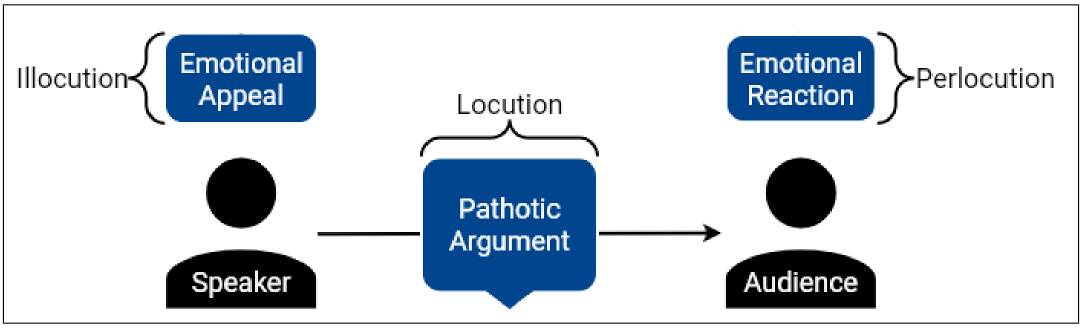

Figure 2: MIPA [21] takes an interactional view of pathos. Redrawn based on figure from Konat (2025) [21]

guments can then be annotated with scores along each of the dimensions of the AQ taxonomy. The AQ taxonomy is not intended to be a complete taxonomy of argument quality, nor is it assumed that every dimension of the AQ taxonomy is relevant for all arguments in any context. Other taxonomical approaches have drawn on other theoretical views, such as deliberative quality [29].

In this paper, I focus on one kind of argument: the *pathotic argument*, and its corresponding argument quality: *pathotic argument quality*. A pathotic argument is an argument that involves the use of *pathos*. Pathos (the arousal of emotions as a rhetorical technique) has been discussed since the works of pre-Socratic Sophist philosophers [19]. Aristotle's treatise on rhetoric [1] is the first to properly systematize the role of pathos as a means of persuasion. Classically, emotional arguments have been studied as fallacies, primarily as fallacies of relevance. More recently, several scholars have challenged traditional views of pathos, asserting that some uses of pathos are relevant, rational, and carry informational content [3, 11, 12, 33].

As with AQA, attention has turned to how computational approaches could automatically detect uses of pathos [8, 13]. The computational mining of pathos has potentially useful applications for studying polarization, and the kinds of arguments used to debate polarizing issues [9, 10]. The Model of Interactional Pathos in Argumentation (MIPA) [21] has been used for the computational analysis of pathos. MIPA views pathos as an interactional event between a speaker and an audience, where an emotional appeal is used in a persuasive setting by the speaker with the aim of eliciting emotions in the audience to increase the (perceived) strength of their arguments. In this paper, I propose that a theory of pathotic argument quality could be added to MIPA to perform pathotic AQA.

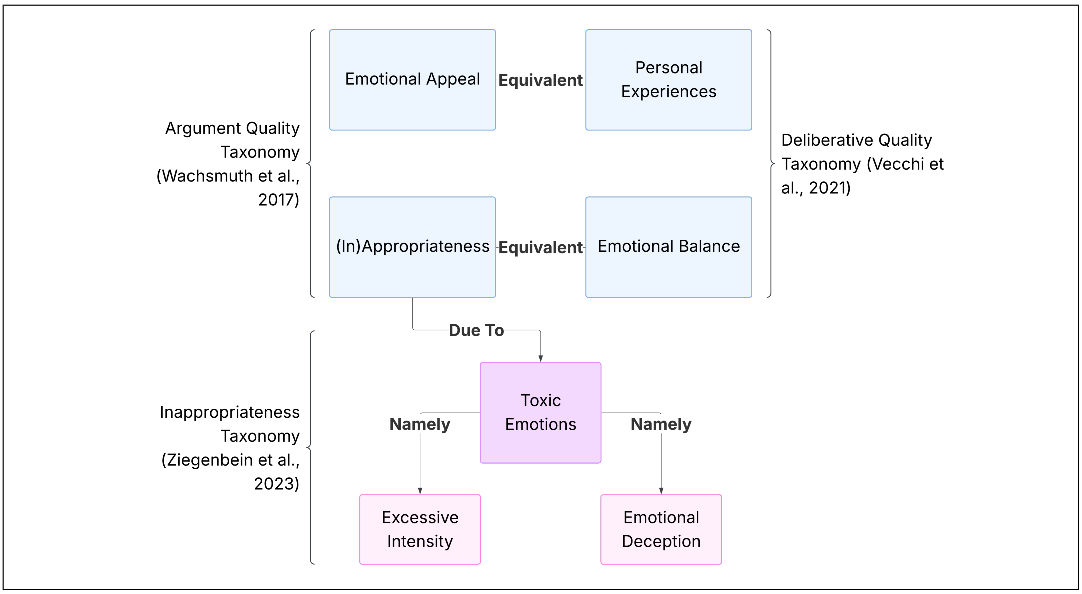

Figure 3: Links between the emotional dimensions of different argument quality taxonomies.

2 Pathotic Argument Quality

MIPA [21] takes an interactional view of pathos, grounded in speech act theory [2]: the speaker has an illocutionary intent to elicit emotion in the audience, and they deliver a locution with emotion-eliciting wording where neutral phrasing could have been used. If the speaker is successful the perlocutionary effect is that the intended emotion is aroused in the audience which increases the perceived strength of the speaker's argument (see Figure 2). MIPA is a promising model for automatic pathotic AQA as it has already been used for several experimental studies [22, 20, 25]. How can we provide an account of pathotic argument quality that meshes with MIPA's description of pathos? MIPA is an interactional model of pathos: it analyses pathos as the interaction between a speaker's argument and an audience's reaction. Therefore, we need a theory of pathotic argument quality that is also interactional. My proposed theory has three dimensions that must be assessed to perform pathotic AQA: a pathotic argument should be *effective*, it should create *pathotic assent*, and it must not be *fallacious*. These dimensions capture the emotional effect, the emotional ground, and the dialectical legitimacy of a good pathotic argument. In Section 3 I discuss how these dimensions of pathotic argument quality could be assessed automatically within MIPA.

2.1 Effectiveness

Most existing approaches to automatically assess pathotic argument quality explore effectiveness: how the use of emotion persuades an audience of the speaker's stance. Assessing the effectiveness of pathotic arguments is closely related explorations of the relationship between uses of emotion and argument convincingness [5, 15, 16].

The previously mentioned AQ taxonomy (Figure 1) captures the impact of pathos on argumentation quality through two sub-dimensions of effectiveness: emotional appeal and appropriateness. The former assesses how an emotional appeal makes an audience more open to the author's arguments; the latter assesses whether argumentation has an appropriate style of language and the right amount of emotional content for its context and audience. Two other argument quality taxonomies include emotional dimensions (shown in Figure 3), however the dimensions of these two taxonomies are equivalent to or subdivide the existing emotional dimensions in the AQ taxonomy.

Effectiveness is already a part of MIPA. In MIPA, the illocutionary intent of the speaker is to elicit an emotion in the audience for the purpose of persuasive gain [21]. The speaker must make intentional choices in their use of emotion-eliciting language to create this effect through their locution. In the interactional setting of MIPA, the effectiveness of the pathotic argument of the speaker should be judged based on the audience's perceived strength of the argument. A "good" pathotic argument should be effective.

2.2 Pathotic Assent

Is an increase in the perceived strength of the speaker's argument all that determines the pathotic argument quality? It doesn't appear so. An interaction between the intended emotion elicited in the speaker's language and the expressed emotion from the audience must take place. Namely, the increased perceived strength of the speaker's argument should take place *due to the arousal of emotions* that the speaker intends to elicit. I call this second component of pathotic argument quality *pathotic assent.*

I was exposed to the idea of pathotic assent by Jamie Dow's the excellent analysis of passions in Aristotle's *Rhetoric* [7]. Although Dow doesn't explicitly name pathotic assent, it forms a key part of his view of pathotic arguments as a kind of proof (*pisteis*). Dow's work is motivated by a seemingly obvious contradiction in Aristotle's treatment of emotion in *Rhetoric.* On the one hand Aristotle is dismissive of the pre-Socratic Sophistic handbook writers [19] and their attempts to arouse emotion in a crowd for persuasive gain, but Aristotle also situates pathos alongside other forms of technical proof (*entechnoi pisteis*): *logos* and *ethos*. Dow argues that Aristotle viewed technical proofs as providing "proper grounds for conviction" [7, p. 34] for the audience. For Dow, a technical proof provides proper grounds for conviction if it can answer two tests:

> 1. Can this thing be cited by believers themselves as a justification for their believing the conclusion?
> 2. Can we cite this thing as a third-party explanation for how a believer was justified in believing the conclusion?
>
> [7, p. 102]

Logos-based and ethos-based proofs are able to answer these tests easily: for logos-based proofs we have proper grounds for conviction because we can cite the argument as justification for our beliefs, and in ethos-based proofs we can cite the character of the authority and their trustworthiness. Pathos-based proofs present more of a challenge, our emotional state does not seem citable as good grounds for conviction in either of the two tests. If we are asked "Why did you think Smith guilty?" and we answer "Because I was scared of him" this is not good grounds for the conviction of our belief [7, p. 103]. Instead, Dow contends that pathos-based proofs are possible if we make the contents of our emotional states (their representation) explicit and cite them as grounds for our belief. In the case of Smith, Dow suggests we do this by using the term "I feel..." to form a propositional attitude: "I believe that Smith is guilty of assault because I feel he is a dangerous character" [7, p.104-105].

Dow's notion of pathos-based proof is slightly too strong for the empirically focused concept of pathotic argument quality I am developing. It seems unlikely that the audience will always directly cite the contents of their emotional states as reasons for their belief. However, from Dow's concept of pathos-based proof we can derive a concept of *pathotic assent* that is compatible with MIPA: the speaker intends to put the audience into a particular emotional state with certain content (fear and "Smith is a dangerous character") and the audience *indicates* that they are in this emotional state (expresses fear and states "Smith is guilty of assault because I feel he is a dangerous character"). I use the term *indicates* to allow pathotic assent to be expressed in degrees. We might observe strong pathotic assent if the audience directly cites their emotional state, and it is the emotional state the speaker intended to put the audience in, or we might observe weaker pathotic assent if the audience makes more indirect references to their emotional state.

A pathotic argument that is both effective and creates pathotic assent in the audience is intuitively of a higher pathotic argument quality than a pathotic argument that is *only* effective, or *only* creates pathotic assent. Consider a lawyer using fear-eliciting language about Smith. The audience may agree that Smith is a dangerous character (effectiveness) while denying that fear played any role in their judgment. Conversely, an audience may express fear in response to fear-eliciting language, without endorsing that Smith is a dangerous character.

2.3 Fallaciousness

The combination of effectiveness and pathotic assent capture the interactional aspects of pathotic argument quality: a pathotic argument should increase the perceived strength of an argument for the audience, and it should do so through the affirmation of the intended emotional content by the audience. However, both effectiveness and pathotic assent are a kind of maximal quality, describing what a pathotic argument should *ideally* be. We might also want to consider relevant notions of minimal quality for pathotic arguments, what a pathotic argument should *avoid*.

Incorporating minimal assessment for pathotic AQA is challenging due to the traditional view that appeals to emotion are fallacies. If all appeals to emotion are fallacious, then pathotic AQA is uninteresting. Walton provides analysis of when emotional appeal is minimally acceptable or fallacious. In his view the fallaciousness of an emotional appeal arises when it illicitly shifts the agreed dialogue type to another dialogue type:

> Emotional appeals and the personalization of arguments are not necessarily impediments or defects in a critical discussion. ... Fallacies often arise where there has been a dialectical shift, a change from one context of dialogue to another. [33, p. 23]

Walton's characterization of fallacies has several desirable properties in relation to pathotic argument quality. Firstly, it prevents uses of pathotic arguments from being intrinsically fallacious. In adversarial dialogue types, such as quarrels, the use of an emotionally charged *ad hominem* is entirely reasonable and expected. Secondly, Walton's view still imposes a firm restriction on the place of emotion in argument, which allows for minimal AQA. If an emotional appeal shifts a scientific inquiry into a critical discussion without the agreement of both parties, then it is fallacious. For example, if Smith is lecturing at a conference on the function of emotional arguments in a courtroom, and someone challenges him because he has previous criminal convictions, this would mark an illicit shift in dialogue type. Finally, focusing on dialogue types also provides a convenient separation of fallaciousness from effectiveness and pathotic assent. Walton notes that the "seeming validity" [33, p. 23] of some fallacious arguments is explained by them being *illicit*

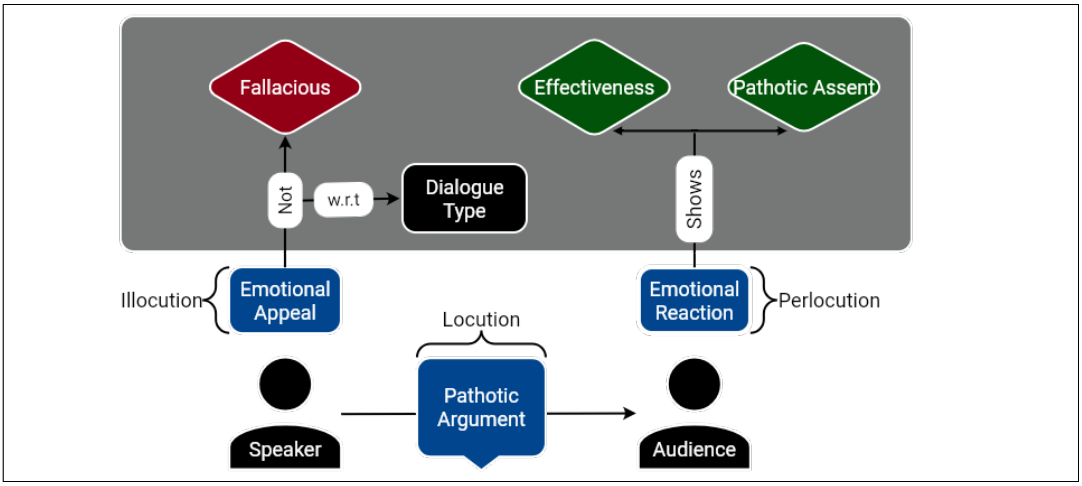

Figure 4: Assessing pathotic argument quality in MIPA.

shifts in dialogue type. A pathotic argument can be effective, and create pathotic assent, yet still be fallacious because the shift in dialogue type is illicit.

I propose that Walton's view should be incorporated into pathotic AQA. Effectiveness and pathotic assent outline what a pathotic argument should do, but it is important to specify what a pathotic argument should avoid doing. One challenge of incorporating fallaciousness into MIPA is that it seems doubtful that we can perform an assessment of fallaciousness based on the interaction between the speaker and the audience. We cannot expect the audience to always identify fallacies themselves: the illicit shift might make the fallacious argument seemingly valid to the audience. Instead, I suggest making the dialogue type an explicit part of MIPA, so that Walton's minimal assessment is performed by comparing the speaker's illocutionary intent to the agreed-upon dialogue type.

3 Assessing Pathotic Argument Quality

To reiterate, a pathotic argument should be *effective*, it should create *pathotic assent*, and it must not be *fallacious*. These dimensions can be incorporated into MIPA, seen in Figure 4. The speaker's emotional appeal should not be fallacious with respect to the dialogue type, and the audience's emotional reaction should indicate that the pathotic argument was effective and created pathotic assent.

One challenge to my proposed theory of pathotic argument quality is to ask why *just* these dimensions. The answer is partly practical. Effectiveness, pathotic assent, and fallaciousness, are all key parts of pathotic argument quality, all fit into MIPA, and some development of computational methods has happened for each. A second challenge is to ask why the dimensions I proposed cannot be combined, especially as pathotic assent and effectiveness appear to be tightly connected. However, distinguishing effectiveness, pathotic assent, and fallaciousness allows us to identify not only persuasive emotional arguments, but also cases of emotionally grounded and emotionally effective yet dialectically illegitimate persuasion. I will not impose that there is one way to determine effectiveness, pathotic assent, and fallaciousness. In a brief survey I review potential approaches to computationally assess each of these dimensions.

3.1 Computational Assessment of Effectiveness

The effectiveness of a pathotic argument in MIPA can be assessed by choosing a measure of effectiveness that is dependent on audience reaction. The chosen measure of effectiveness will depend on the context, and the available data. Here I highlight two example approaches that perform automatic pathotic AQA by observing audience reactions.

Cano-Basave & He (2016) [4] study the impact of persuasive argumentation in political debates. They extract several linguistic features from the arguments in presidential debates to see how well these features predict how influential a speaker is on the audience. The speaker's influence is measured by their standing in polls taken before and after the debate. An increase in a speaker's influence indicates effectiveness in the speaker's arguments. Two of the features are emotional: a set of generated emotion-related semantic frames on the part of the speaker, and non-linguistic external emotion cues (booing and cheering from the audience) from the audience. Cano-Basave & He find that emotional content is a strong indicator of the effectiveness of the speaker's arguments.

Lukin et al. (2017) [24] present a set of socio-political arguments to a group of subjects. The subjects are profiled for personality traits and prior beliefs. Lukin et al. then evaluate arguments by observing what degree of belief change can be observed in the subjects before and after being presented with an argument, and trying to predict degree of belief change. The collected arguments are sorted into categories, including factual (non-pathotic), emotional (pathotic). They found that all types of arguments were effective in triggering belief change, with factual and emotional being equivalent in strength. In predicting whether an argument was effective for a subject, they observed consistently better results when incorporating a subject's personality traits.

3.2 Computational Assessment of Pathotic Assent

I propose that the pathotic assent of a pathotic argument in MIPA can be assessed by observing if the audience indicates that they affirm the emotional content the speaker tries to elicit. At a coarse-grained level, the primary emotion the speaker is trying to elicit in their pathotic argument can be identified, and the primary emotion of the audience's reaction can be identified, and a comparison can be performed. As far as I am aware only a coarse-grained assessment of pathotic assent has been performed. I describe the coarse-grained approach below, and then move onto to discussing potential methods for automatic assessment of pathotic assent.

Konat et al. (2024) [22] collect transcripts of political debates, along with collect online social media reactions (tweets) from the same period. The emotional content of the speakers' arguments are collected by annotating debate arguments with their employed argument schemes and identifying emotion-eliciting words. To analyze audiences reactions, tweets are classified by their expressed emotion. Konat et al. then organize audience reactions into those that occurred at the same time as the speakers' arguments. They find that, of the various emotions they analyze, politicians are generally successful at inducing negative emotions in their audience through arguments that can be identified as negative pathotic arguments.

Their approach captures a coarse-grained view of pathotic assent. It also relies on the immediate audience responses to the pathotic argument to convey that pathotic assent; this is sensible as Konat et al. aim to analyze the immediate social media reactions that co-occur with pathotic arguments delivered in a debate. However, in conversational contexts, such as debates in online forums, audience reactions could be tracked over a longer period of time. For example, beyond just observing individual emotional reactions from the audience in response to an argument, the emotional dynamics [18] of the audience can be tracked. Emotional dynamics measures the baseline use of different emotional language of an audience, and then observes displacements in an audience from these baselines. In an argumentative context, displacements in emotion could indicate strong emotional reactions to arguments or claims.

To determine causal relationships between pathotic arguments and emotional reactions I suggest that attention should be paired to the relatively recent computational task of emotion-cause pair extraction in conversations (ECPEC) [23]. ECPEC aims to extract emotional utterances from a conversation and the associated utterances that caused the emotion (see Figure 5). Crucially in ECPEC utterances can cause their own emotions (or rather the expressed emotions are not caused by other utterances), which could allow for cases of genuine causally related pathotic assent to be distinguished from cases where the audience just expressed emotion inde-

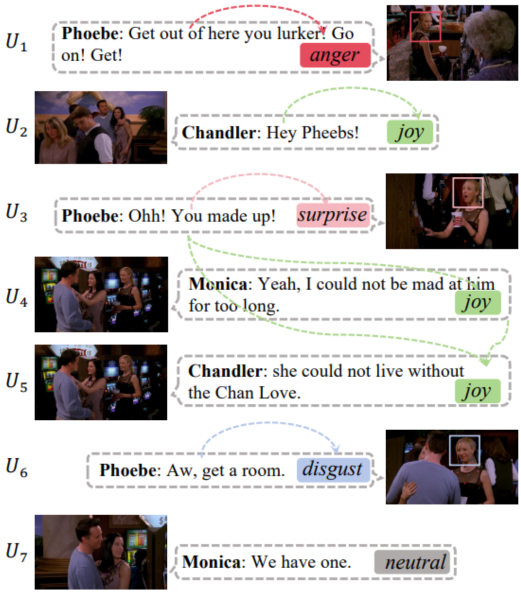

Figure 5: Example of emotion-cause pairs in conversation at the utterance level, dashed lines represent emotion causes between utterances. Figure from Cheng et al. 2024 [6], licensed under the Creative Commons Attribution 4.0 International License (CC BY 4.0), `https://creativecommons.org/licenses/by/4.0/`

pendently of any arguments made. As far as I am aware ECPEC has never been applied to argument mining or computational argumentation. A barrier for applying ECPEC to argument mining is the relatively small number of datasets, especially since no existing ECPEC datasets are curated with argument mining in mind.

3.3 Computational Assessment of Fallaciousness

A good pathotic argument should not be fallacious. I have outlined that in my view, a proper treatment of the fallaciousness of pathotic arguments should be rooted in a Waltonian approach, allowing for emotional appeals to be fallacious not intrinsically but rather because they are an illicit shift or move in a discussion with respect to a dialogue type.

The computational assessment of fallacies has received considerable attention, and numerous datasets to support fallacy detection have been introduced (see Ramponi et al. (2025) [26] for an overview). However, the underlying theory of fallacies within approaches and existing datasets varies. For example, the MAFALDA dataset

[17] contains arguments from a number of datasets annotated according to a taxonomy of fallacies, including a number of 'appeal to emotion fallacies'. However, the annotation guidelines for 'appeal to emotion fallacies' (such as 'appeal to pity') do not refer to the dialogue type, and instead rely solely on the structure and content of the argument.

As far as I am aware, a method of automatic fallacy detection based on illicit shifts in dialogue types has not been introduced. The lack of computational work here presents a potential gap in automatic AQA.

4 Conclusion

A good pathotic argument is *effective*, creates *pathotic assent*, and is not *fallacious*. By separating out these dimensions of pathotic argument quality, pathotic AQA can be formed with greater granularity. These dimensions of pathotic argument quality can be added to the Model of Interactional Pathos in Argumentation (MIPA) [21], to perform pathotic AQA. Additionally, I have outlined existing and potential methods for the computational assessment of automatic pathotic AQA. In future, I plan to undertake some of this computational work myself, focusing on methods for the automatic assessment of pathotic assent.

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